

WASP-12b

R-0260

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241106 · created 2026-07-17T08:51:44+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460620.927 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.0077
Transit depth (Rp/Rs)^2	2.28 +/- 0.23 [%]
Orbital Inclination (inc)	81.85 +/- 2.04
Airmass coefficient 1 (a1)	1.0334 +/- 0.0074
Airmass coefficient 2 (a2)	-0.028 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	#2 - [493, 25]
Optimal Aperture	3.81
Optimal Annulus	7.02
Transit Duration (day)	0.1254 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.0077 vs 0.1178 (deviation 4.31σ)
Duration fit vs published	0.1254 d vs 0.125 d (deviation 0.08σ)
Mid-time O–C	2.2 min
Depth SNR	19.6
Residual scatter	0.72 %

Light curve

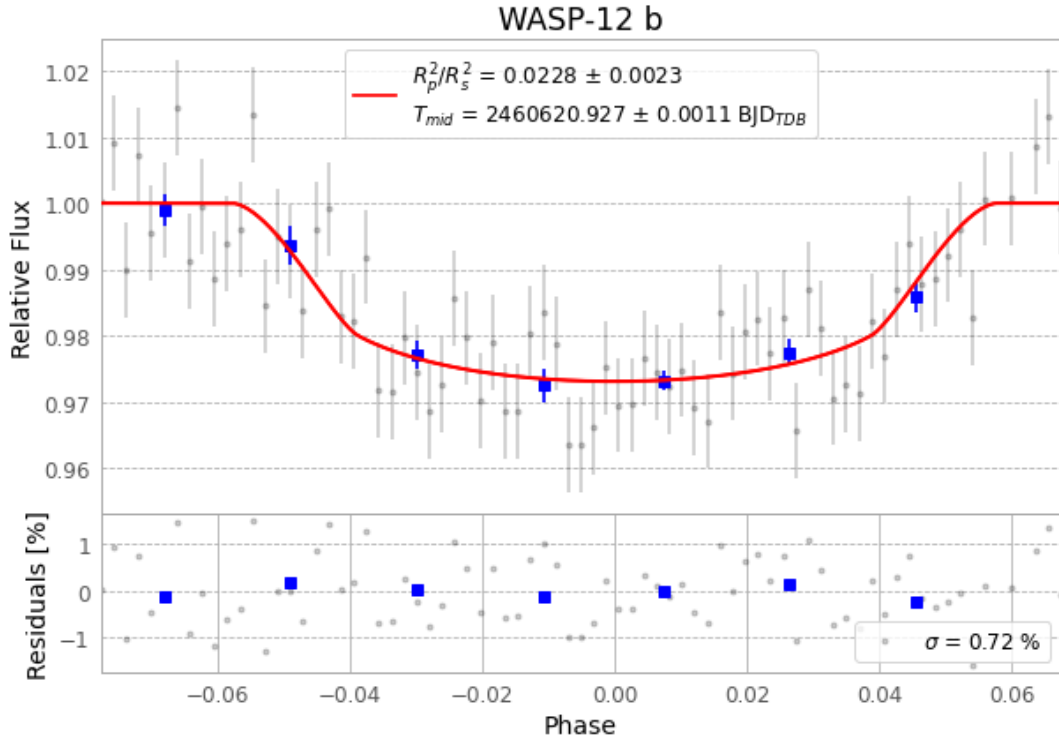


Figure 1 — FinalLightCurve_WASP-12 b_2024-11-06.png (SHA-256 0013767ecb9f6ee0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2db7717667ce16d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 13bb8b5

Data provenance

77 FITS frames (51 MB), WASP-12241106080616.FITS → WASP-12241106115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241106.log (8f3a21f71f7f...), FinalLightCurve_WASP-12 b_2024-11-06.png (0013767ecb9f...), FinalLightCurve_WASP-12 b_2024-11-06.csv (3efb881d6f55...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 08:51Z.